

Automated Multi-Branch Enterprise Network

by

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UNDER THE SUPERVISION OF

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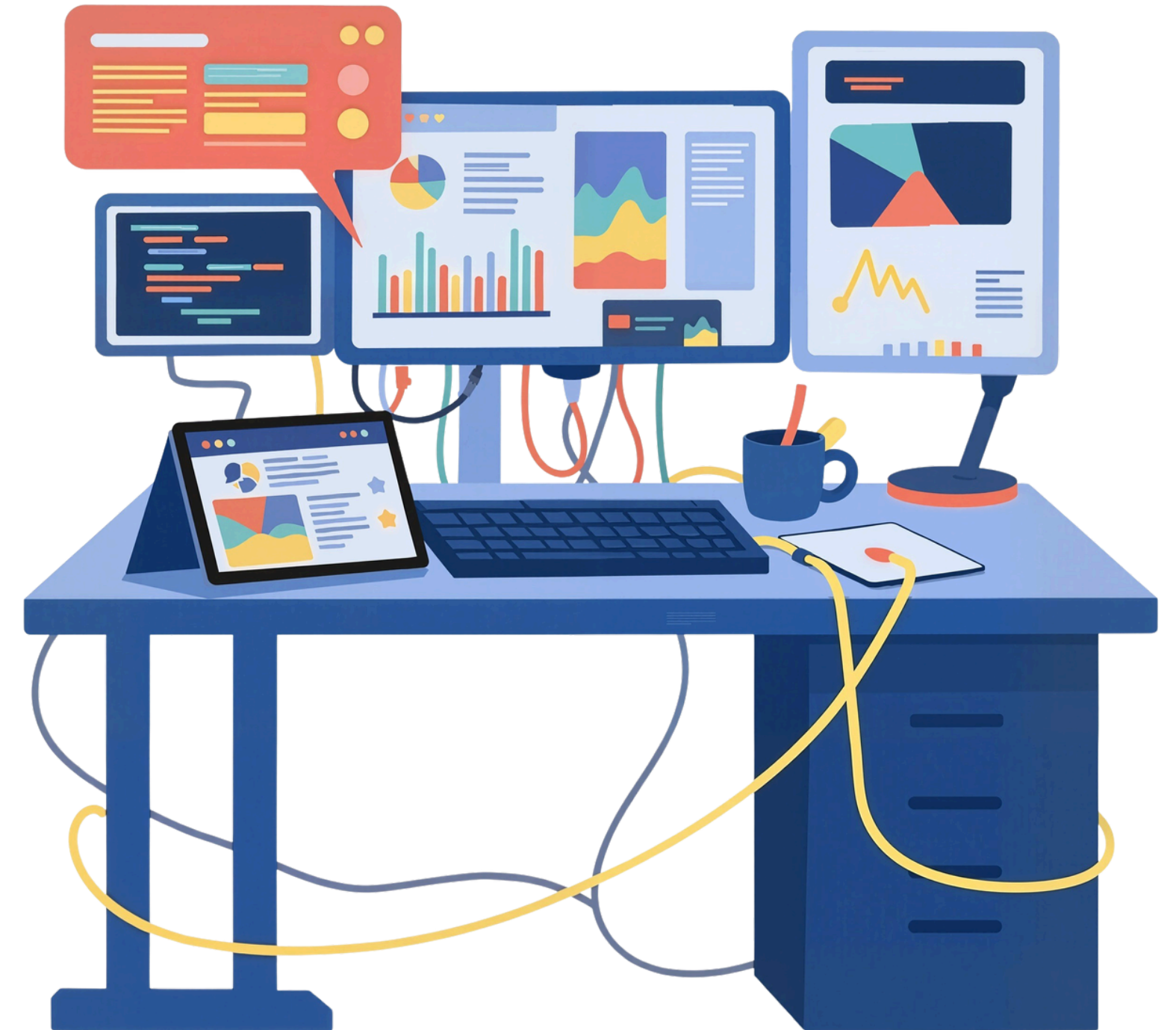
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Table of Contents

- Problem Statement
- Project Objective
- Project Methodology
- System Overview
- Network Architecture
- Technology and Protocols Used
- IP Addressing
- Algorithm Implementation
- Test Cases and Results
- Conclusion



Problem Statement

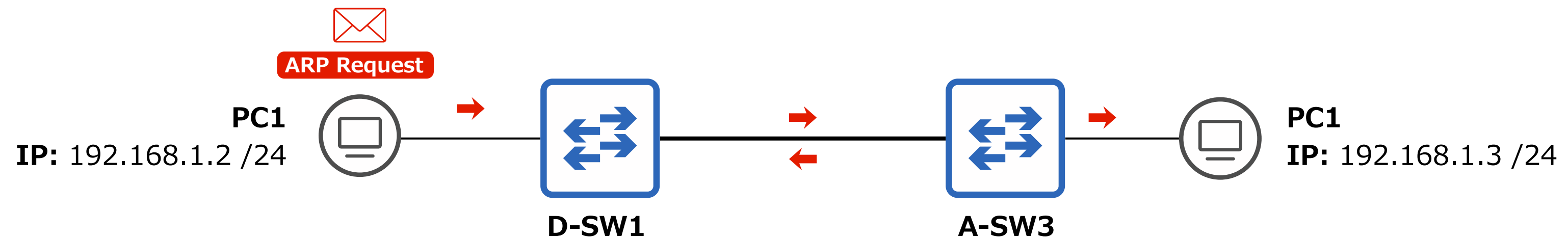


Source: Cisco Systems, <https://www.cisco.com>

- Manual network management
- No centralized monitoring
- Security risks across branches
- No automatic failover
- Difficult troubleshooting
- Lack of automation

Project Objectives

- Design a scalable and highly available multi-branch network.
- Implement secure WAN connectivity with centralized monitoring.
- Automate network configuration and management



Project Methodology

Cisco **PPDIOO** Lifecycle Model



Prepare → Requirement analysis

Plan → Feasibility study & resource planning

Design → Network topology & IP addressing

Implement → Configure, automate & dashboard

Operate → Monitoring, syslog & health scoring

Optimize → Performance & failover improvement

System Overview

- Enterprise Multi-Branch Network
- Dynamic Routing & Secure WAN
- Centralized Linux Automation Server
- Python-Based Automation
- Centralized MySQL Database
- Real-Time NOC Dashboard

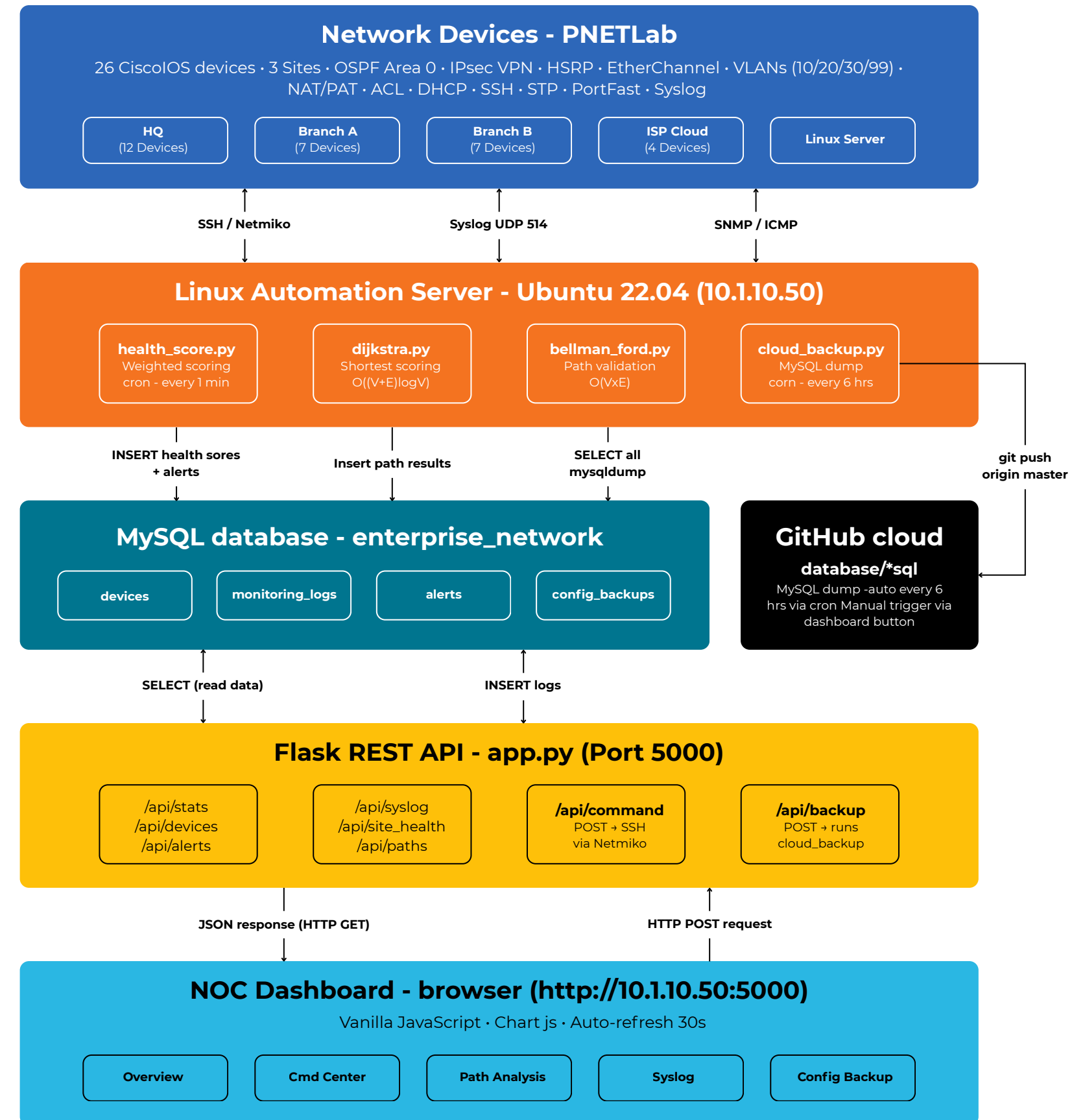


Fig : System Architecture

Network Architecture

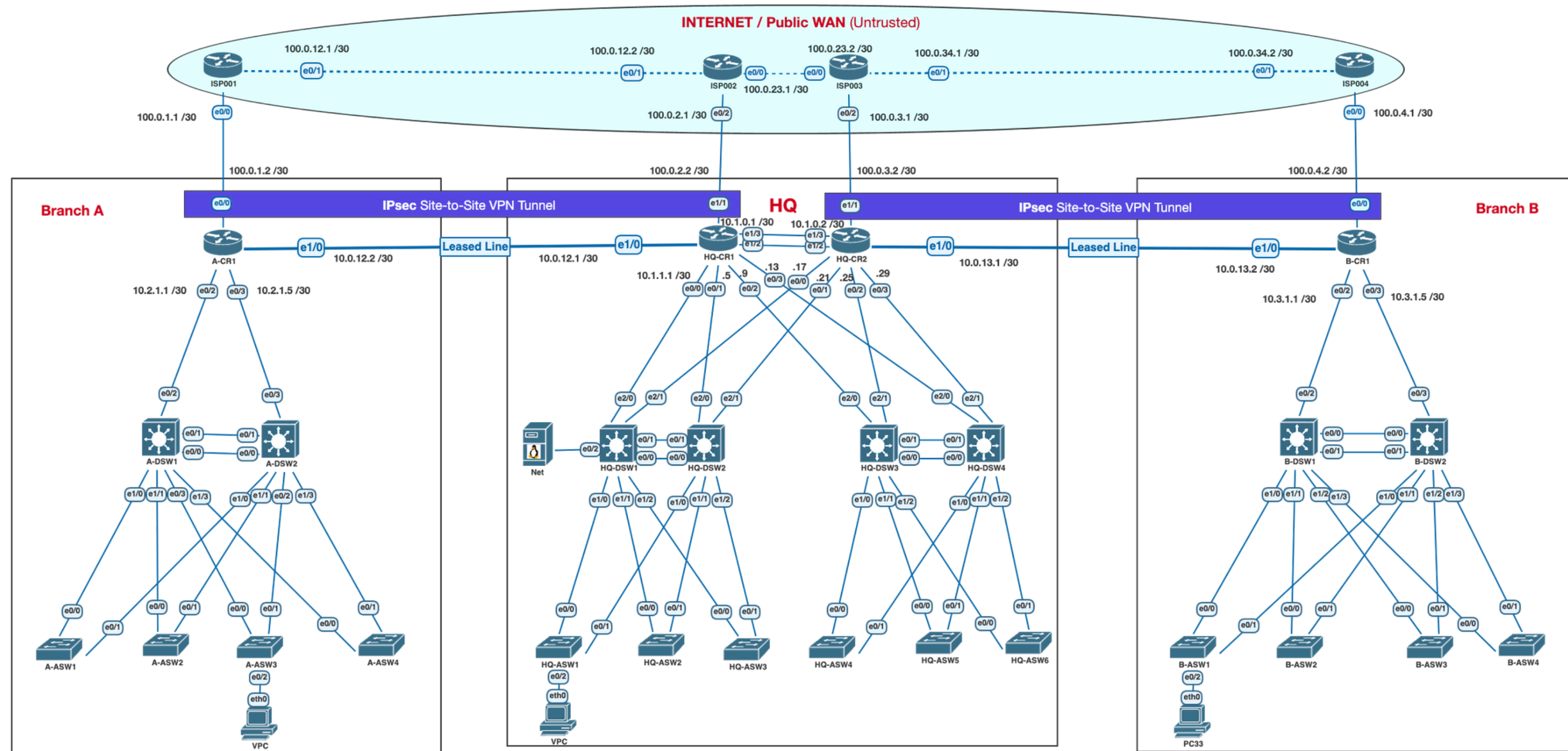


Fig: Logical Network Topology Diagram

Technology & Protocols Used

Network Protocols

- OSPF Area 0
- IPsec VPN (AES-256)
- HSRP v2
- Etherchannel LACP
- 802.1Q Trunking
- DHCP
- SSH v2
- STP Rapid-PVST
- PortFast
- Extended ACL

Automation

- Python 3.10
- Flask
- Netmiko (SSH)
- Linux Cron
- Git / Github
- REST API
- PyMySQL
- Bash Scripting

Monitoring

- MySQL 8.0
- rsyslog (UDP 514)
- SNMP (UDP 161)
- ICMP Ping
- Chart.js
- Vanilla Javascript

Tools & Platform

- PNetLab (Emulation)
- Cisco IOS vIOS
- Ubuntu Server 22.04
- Vmware Workstation
- MobaXterm
- Wireshark
- Microsoft Word
- Draw.io
- MS Project

IP Addressing

Table: Network Address Allocation Overview

Site	Address Block	Subnet Mask	No. of Devices
Headquarters (HQ)	10.1.0.0 / 16	255.255.0.0	12 devices
Branch A	10.2.0.0 / 16	255.255.0.0	7 devices
Branch B	10.3.0.0 / 16	255.255.0.0	7 devices
ISP Cloud (Public)	100.0.0.0 / 8	255.0.0.0	4 ISP routers
Leased Line HQ↔A	10.0.12.0 / 30	255.255.255.252	Point-to-point
Leased Line HQ↔B	10.0.13.0 / 30	255.255.255.252	Point-to-point

Algorithm Implementation

1) Dijkstra's Algorithm

- Approach: Greedy • Min-heap priority queue
- Complexity: $O((V+E)\log V)$

$$O((V + E)\log V) = O((12 + 30)\log 12) \approx O(176)$$

2) Bellman-Ford Algorithm

- Approach: Dynamic programming
- Complexity: $O(V \times E)$

$$O(V \times E) = O(12 \times 30) = O(360)$$

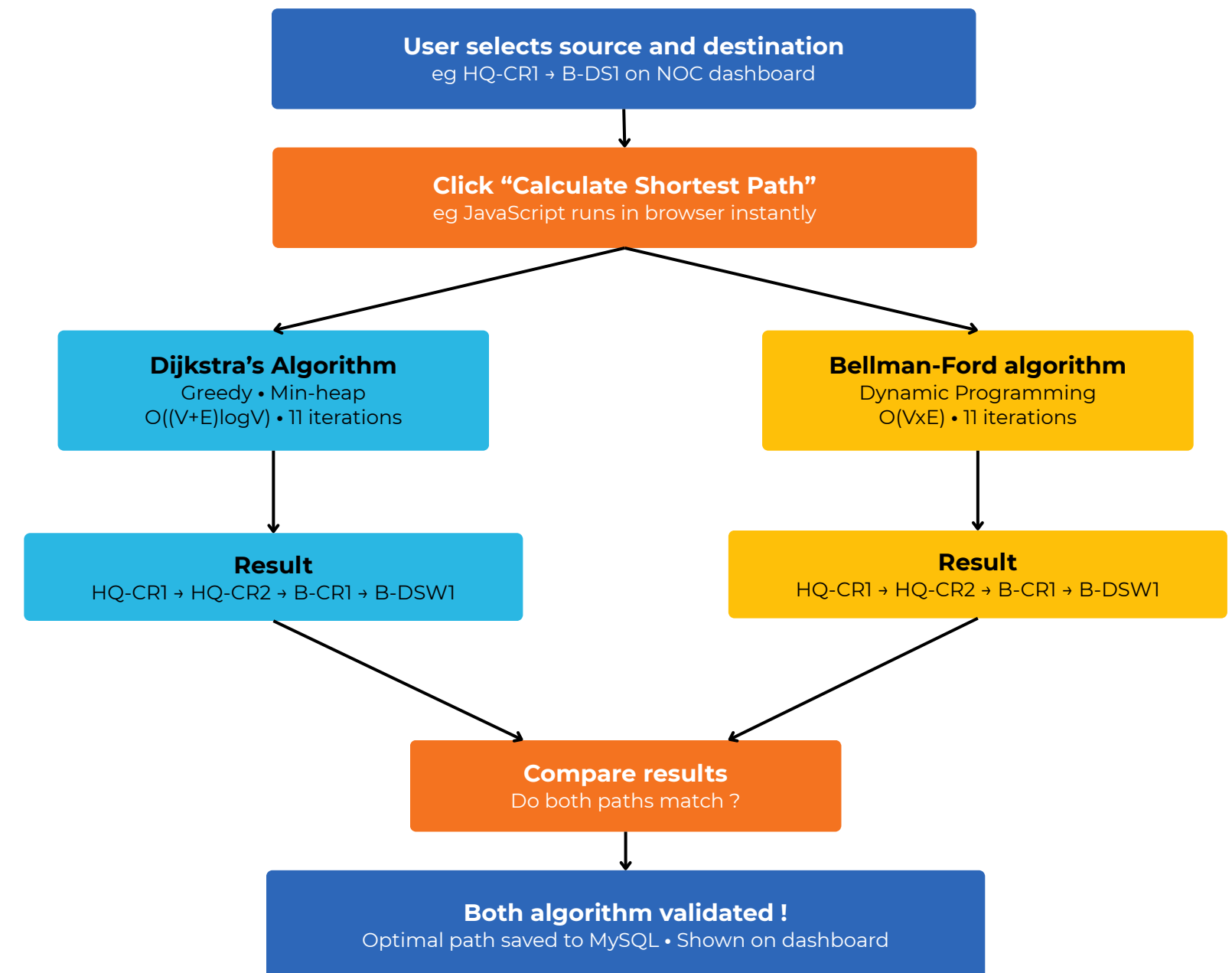


Figure: NOC Dashboard Path Analysis: Dijkstra and Bellman-Ford Algorithm Execution Flow and Cross-validation Process

Algorithm Implementation (Contd)

3) Weighted Health Scoring Algorithm

- Approach: ICMP-based weighted scoring
- Complexity: O(n)
- Trigger: Linux cron
- Score = Connectivity + Response Time
- Storage: MySQL monitoring_logs + alerts table

$$\text{Response Time Score} = 40 \times \max \left(0, 1 - \frac{\text{avg}_{rtt}}{100} \right)$$

$$\text{Health Score} = \text{Connectivity Score} + \text{Response Time Score}$$

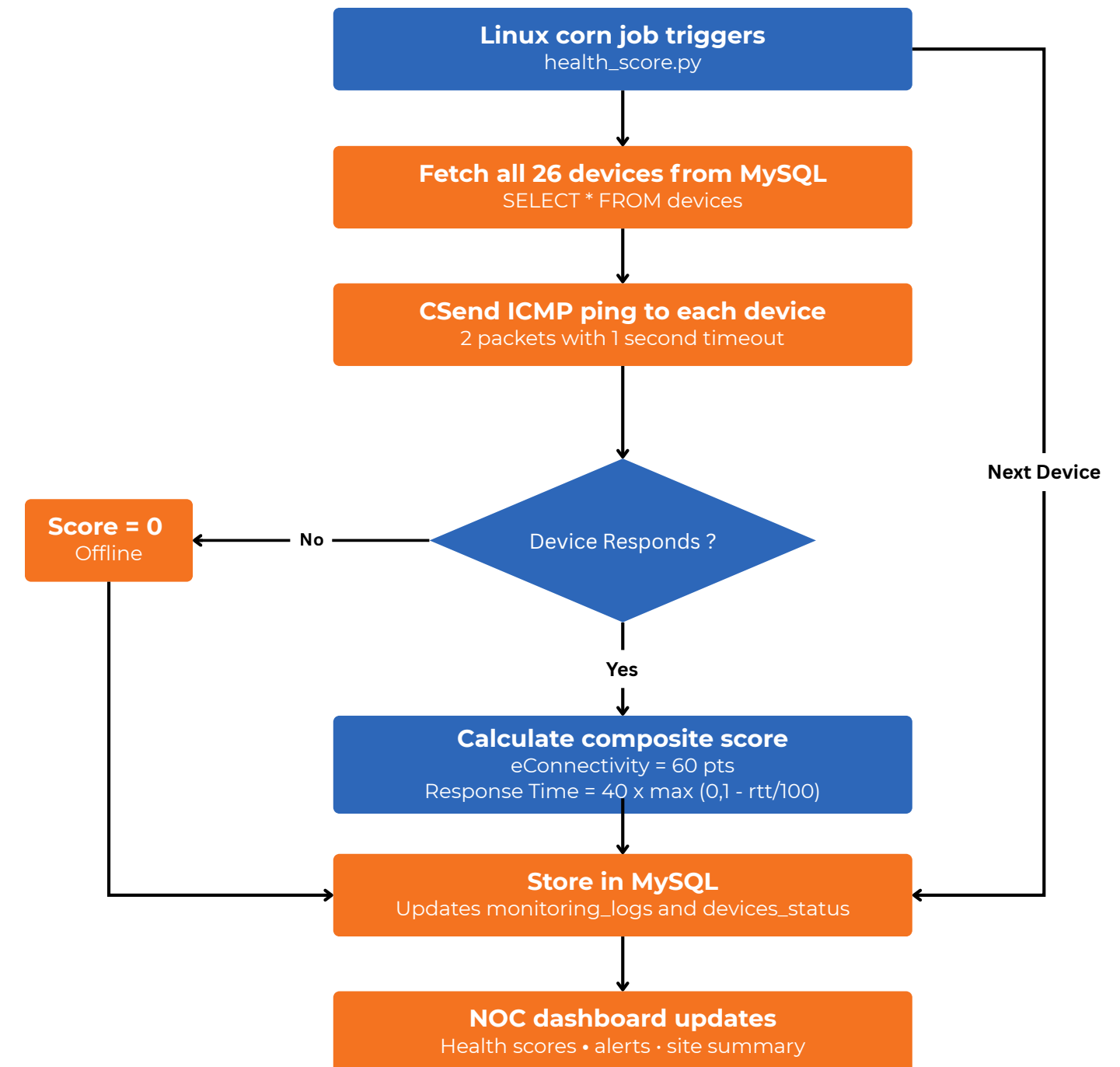


Figure: Weighted Health Scoring Algorithm for Automated Device Monitoring.

Test Cases and Results

Table: Connectivity Testing Results

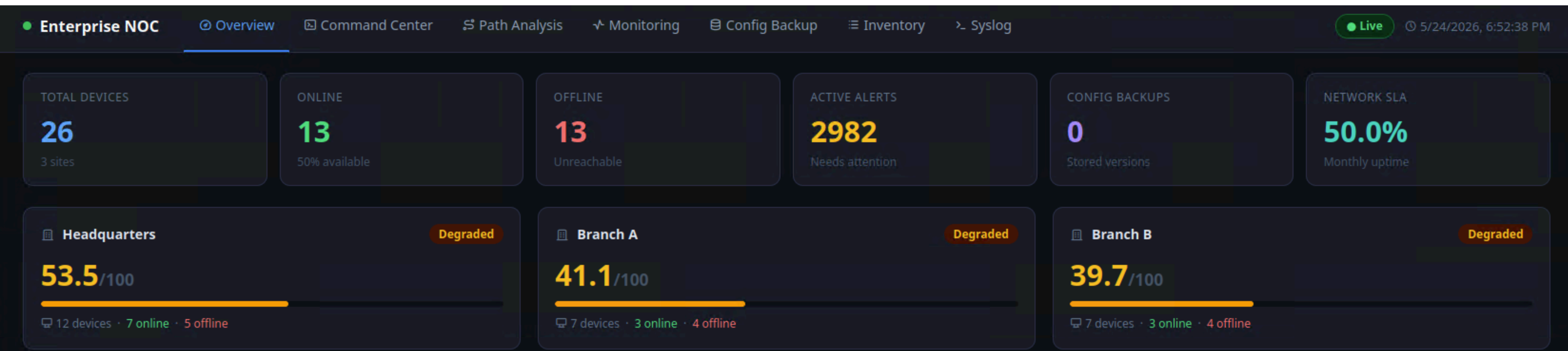
Test ID	Test Case	Expected	Actual Result	Status
TC-01	HQ to Branch A ping	Success via leased line	100% success, <7ms RTT	PASS
TC-02	HQ to Branch B ping	Success via leased line	100% success, <6ms RTT	PASS
TC-03	Branch A to Branch B	Blocked by ACL	Unreachable as expected	PASS
TC-04	OSPF neighbor adjacency	FULL state on all links	All neighbors FULL/DR	PASS
TC-05	DHCP address assignment	FULL state on all links	All hosts assigned correctly	PASS
TC-06	SSH management access	FULL state on all links	Admin login verified	PASS

Test Cases and Results (Contd.)

Table: VPN Failover Testing Results

Test ID	Test Case	Expected	Actual Result	Status
TC-07	Shutdown leased line HQ↔A	VPN activates automatically	IPsec QM_IDLE ACTIVE within 15s	PASS
TC-08	Ping HQ to Branch A via VPN	Traffic via IPsec tunnel	100% success, 7–12ms RTT	PASS
TC-09	Shutdown leased line HQ↔B	VPN activates for Branch B	IPsec SA active, B-CR1 reachable	PASS
TC-10	Restore leased line	Traffic returns to leased line	OSPF reconverges, VPN tears down	PASS
TC-11	VPN encryption verify	AES-256 encrypted packets	show crypto ipsec sa confirms	PASS

Conclusion



- Successfully built and tested a secure multi-branch enterprise network.
- Achieved centralized monitoring, automation, and reliable connectivity.
- Developed a real-time dashboard for easier network management.
- Demonstrated a scalable and cost-effective enterprise networking solution.

Thank You

We sincerely appreciate your time and
consideration

Questions & Answers